4.2

PAEDIATRICS

Cardiovascular System Infections

CHAPTER 4.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Infective endocarditis			
Viridans Streptococci and other Streptococci spp			
Native valve, Non IVDU, prosthetic valve *Strep viridans* *Other Streptococci*	**<u>MIC < 0.125ug/ml</u>** Benzylpenicillin 50,000 units/kg/dose IV q6H for 4 weeks (6 weeks for prosthetic) **<u>MIC > 0.125 to 2 ug/ml</u>** **ADD** Gentamicin 1 mg/kg/dose IV q8H for 2 weeks (6 weeks for prosthetic)	**<u>MIC < 0.125ug/ml</u>** Ampicillin 75 mg/kg/dose IV q6H for 4 weeks (6 weeks for prosthetic) **OR** Ceftriaxone 100 mg/kg/dose IV q24H for 4 weeks (6 weeks for prosthetic) **<u>MIC > 0.125 to 2 ug/ml</u>** **ADD** Gentamicin 1 mg/kg/dose IV q8H for 2	• At least 3 sets of blood cultures from different sites before empiric therapy. • **<u>Penicillin allergy:</u>** Vancomycin 15 mg/kg/dose IV q8H for 4 weeks (6 weeks for prosthetic) **PLUS** Gentamicin 1 mg/kg IV q8H for 2 weeks (6 weeks for prosthetic)

CHAPTER 4.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
		weeks (6 weeks for prosthetic)	
Staphylococcus aureus			
Native valves			
Left sided endocarditis *Staphylococcus aureus* (MSSA, MRSA)	**For MSSA IE:** Cloxacillin 50 mg/kg/ dose IV q6H for 4-6 weeks **For MRSA IE:** Vancomycin 15 mg/kg/ dose IV q8H for 4 - 6 weeks	**For MSSA IE:** Cefazolin 30 mg/kg/dose IV q8H for 4-6 weeks	
Right sided endocarditis *Staphylococcus aureus* (MSSA, MRSA)	**For MSSA IE:** Cloxacillin 50 mg/kg/ dose IV q6H for 4 weeks **For MRSA IE:** Vancomycin 15 mg/kg/ dose IV q8H for 4-6 weeks		

CHAPTER 4.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Prosthetic valves			
Methicillin-susceptible *Staphylococcus aureus* (MSSA)	Cloxacillin 50 mg/kg/ dose IV q6H for >6 weeks + Rifampicin 20 mg/kg/day PO q8H for >6 weeks + Gentamicin 1 mg/kg/ dose IV q8H for 2 weeks		• Rifampicin has better penetration. However, to avoid the development of resistance, it should be started after 3-5 days of effective initial cloxacillin therapy and/or once the bacteraemia has been cleared.
Methicillin-resistant *Staphylococcus aureus* (MRSA)	Vancomycin 20 mg/kg/ dose IV q8H for > 6 weeks + Rifampicin 20 mg/kg/day PO q8H for >6 weeks + Gentamicin 1 mg/kg/ dose IV q8H for 2 weeks		• Rifampicin has better penetration. However, to avoid the development of resistance, it should be started after 3-5 days of effective initial vancomycin therapy and/or once the bacteraemia has been cleared.

CHAPTER 4.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Pericarditis (Purulent)			
Staphylococcus aureus *Streptococcus pneumoniae* *Haemophilus influenzae* Other Streptococci Gram-negative bacteria	Cloxacillin 50 mg/kg/ dose IV q6H for 4 - 6 weeks + Gentamicin 1 mg/kg/ dose IV q8H for 2 weeks	Vancomycin 15 mg/kg/ dose IV q8H for 4 - 6 weeks + Gentamicin 1 mg/kg/ dose IV q8H for 2 weeks	• Duration: minimum 4 weeks. • Early surgical consultation for possible intervention. Ceftriaxone / Cefotaxime may be considered in place of Gentamicin.
Rheumatic Fever			
Post infectious sequelae of Group A strep infection	Benzylpenicillin 50,000 units/kg/dose IV q6H for 10 - 14 days Followed by, Penicillin V prophylaxis: • 125 mg q12H (<5 years old) • 250 mg q12H (>5 years old)	**Penicillin Allergy:** Erythromycin 15 mg/kg/dose PO q6H for 10 - 14 days Followed by, Erythromycin 15 mg/kg PO q12H	
Pacemaker / Defibrillator & Ventricular Assist Device Related Infection			

CHAPTER 4.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
S aureus *S epidermidis* Aerobic gram-negative bacilli Fungal	Vancomycin 15 mg/kg/ dose IV q8H for 6 weeks + Gentamicin 1 mg/kg/ dose IV q8H for 2 weeks		• IV to PO switch is possible after clinical improvement. • Fungal infection is rare. Amphotericin B or Voriconazole is equally effective for proven fungal infections.

Infective Endocarditis Prophylaxis Guideline for Dental Procedure
Refer to **Chapter 3.18: Non-surgical Prophylaxis**.